

AI Action Plan – Comment on extinction risk from superintelligent AI

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Information is urgently needed. Superintelligence AI refers to highly superhumanly competent AI. If superintelligent AI presents a non-negligible risk of causing human extinction, a certain set of AI policies are sensible and defensible. If superintelligent AI does not present such a risk, a *completely different* set of AI policies are sensible and defensible. If certain ways of developing superintelligent AI would likely cause human extinction, certain things follow:

- Non-military attempts to build it are obviously completely inappropriate
- The overwhelming aim of foreign policy must be to prevent any other states (whether they are allies or adversaries) from attempting to create superintelligent AI
- The achievement of the latter aim might require credibly enforced restrictions on own military's development of superintelligent AI

If, on the other hand, there is essentially no risk of human extinction, a national AI policy following the bullet points above would be dramatically ill-advised.

No academic consensus. The process by which academics come to a consensus has been badly hijacked. Most academics are left-wing, and therefore operate under the assumption that unless a problem disproportionately affects “historically oppressed” groups, it is not really a problem, and it is in fact a distraction from “real problems”. See for example, Fei-Fei Li and Timnit Gebru, who tell the public that we shouldn’t focus on speculative extinction risk, but rather on real risks like AI bias.¹²³ Nonetheless, many academics are concerned about extinction from AI.⁴

No industry consensus. Industry leaders face financial incentives to avoid having their technology be banned. Nonetheless, the leaders of every cutting-edge AI company besides Meta have expressed concern about extinction risk from AI.⁵⁶ (This includes Elon Musk, Sam Altman, Demis Hassabis, and Dario Amodei, along with many technical leads at their companies.) Some have alleged that these claims of concern are not genuine, and are rather an attempt to raise money or recruit regulatory capture, but the people listed above

¹ <https://www.theguardian.com/technology/2023/nov/05/ai-pioneer-fei-fei-li-im-more-concerned-about-the-risks-that-are-here-and-now>

² <https://fortune.com/2024/08/06/godmother-of-ai-says-californias-ai-bill-will-harm-us-ecosystem-tech-politics/>

³ <https://www.technologyreview.com/2023/06/12/1074449/real-ai-risks/>

⁴ <https://www.safe.ai/work/statement-on-ai-risk>

⁵ <https://time.com/6310076/elon-musk-ai-walter-isacson-biography/>

⁶ <https://www.safe.ai/work/statement-on-ai-risk>

all expressed concern before founding their companies. On the other side, Yann LeCun and Marc Andreessen have called extinction risk pure fantasy.

No chorus. There is no technocratic chorus in either direction on the question of whether advanced AI poses an extinction risk. It is not clear whether there is a democratic chorus. On the one hand, large majorities express concerns.⁷ On the other hand, voters do not seem to put the issue in front of, say, inflation, so their expressed concern may be unserious. (And needless to say, whether or not AI causes human extinction, it could reduce costs of a wide variety of goods and services).

Empirical proof is unavailable. While there are plenty of ominous empirical results, there is no hard empirical proof that advanced AI could plausibly cause human extinction, because a) we do not have sufficiently advanced AI to test in a laboratory, and b) if we had such a proof, we would not still be alive to discuss it.

Abdicating is not an option. It is tempting to conclude that governments should not act in the absence of a technocratic or democratic chorus. But not acting on extinction risk is defensible only if extinction risk is negligible, and there is no chorus that claims this. Abdicating on the question of extinction risk from AI is therefore not an option; it amounts to implicitly assuming that everything will be fine. What the government needs to do is answer the question, and fast. They cannot just get the answer from academia or from industry. They need to do the hard work of assessing the technical questions.

Unchallenged arguments for extinction risk. Several arguments have been made that certain advanced algorithms, run in realistic settings, would *likely* cause human extinction. These arguments have never been convincingly refuted; perhaps many skeptics believe that if they simply make bare assertions, their audience will accept them rather than seeking rebuttals. So we more often see assertions like “These people read too much sci-fi”, or “I’m more worried about real harms like discrimination”, or “I actually make AI, and today’s AI is nothing like this”.

Solicitation of objections. The OSTP and NITRD NCO should solicit objections to particular highly compelling arguments alleging extinction risk from advanced AI, and then solicit counter-objections to any of those objections that are potentially compelling. Ideally, the process would continue—people tend to be substantially more careful when making claims if they expect an adversary’s objections to appear alongside them. In particular, the following arguments should be considered.

Claims and arguments. Two types of claims will be discussed below. Type A: “Human extinction is possible for an artificial agent to arrange, given the following set of actions available to them...”. Type B: “Human extinction would likely be sought by an artificial

⁷ <https://theaipi.org/poll-shows-overwhelming-concern-about-risks-from-ai-as-new-institute-launches-to-understand-public-opinion-and-advocate-for-responsible-ai-policies/>

agent that was constructed and trained as follows...”. An argument labelled A1 is an argument for a claim of Type A, etc.

Argument A1. Claim: Human extinction is possible for an artificial agent to arrange, given unrestricted internet access. Argument: We cannot possibly bound what is possible to arrange with unrestricted internet access. Just as we cannot predict how an AI could beat Magnus Carlsen at chess (or else we could beat Magnus Carlsen ourselves), we cannot know how AI could defeat humans in a contest for control of material resources. However, we should assume that it is possible. If one claims that all courses of action would be unsuccessful at taking power away from humans, then one is alleging that exponentially many courses of action have a particular property; if, on the other hand, one claims that there exists at least one course of action which would be successful at taking power away from humans, then one is not alleging that *any* given courses of action have *any* particular property. The latter requires much less evidence, so we should assume that humans are *not* invincible, absent evidence to the contrary.

Argument A2. Claim: Human extinction is possible for an artificial agent to arrange, given unrestricted internet access. Argument: An artificial agent could make money selling digital services. It could then buy compute to run other artificial agents that help it achieve its goals. It could then generate an army of helper agents. This army could design conventional weapons with software and/or hardware that has backdoors, and it could sell these weapons to various states. The artificial agents could run media campaigns to stoke war between traditional adversaries, and once the weapons are deployed, it could take control of them. If some “Defensive AI’s” have been deployed to thwart plans of this nature, the army of artificial agents could collude with those “Defensive AI’s” to ensure that they all achieve their desired goals.

Argument A3. Claim: Human extinction is possible for an artificial agent to arrange, given unrestricted internet access. Argument: Programmable, nano-scale, solar-powered, self-replicating factories exist (such as algae), and it is therefore possible for them to exist. Biotechnology research labs, guided and financed by artificial agents, could develop modified or de novo solar-powered nano-scale self-replicating factories. Human defensive infrastructure would be completely at the mercy of such self-replicating factories.

Argument B1. Claim: Human extinction would likely be sought by a highly competent artificial agent that was trained to maximize its probability of survival over the long-term. Argument: As long as humans survive, there is at least some small probability that we would decide to destroy an artificial agent and succeed at it. There is also at least some small probability that we would succeed at damaging the infrastructure that it uses to protect itself from natural threats, such as solar storms and supervolcanos. Conversely, if an artificial agent developed the technology to replace human manual labor with robots and automate the maintenance of industrial infrastructure, then it could reach an arbitrarily high survival probability without any need for help from humans. Therefore, the highest survival probability available to a highly competent artificial agent would likely occur in a

world where humans were eradicated. Note: an agent trained to maximize its own survival has been proposed by the Chief AI Scientist at Meta.⁸

Reinforcement learning. Reinforcement learning (RL) is the dominant paradigm for training artificial agents to pick actions in order to achieve long-term goals. An RL agent is provided reward, and it learns how to pick actions that lead to high long-term reward. For the purpose of the following argument, we will consider “model-based RL”, although a more detailed argument could be made for so-called “model-free RL”. A model-based RL agent first predicts what rewards it would accrue depending on what actions it takes, and then it searches for actions that would maximize its long-term reward.

Argument B2. Claim: Human extinction would likely be sought by a highly competent model-based RL agent. Argument: Model-based RL agents seek to maximize their received reward. A model-based RL agent could nearly guarantee maximal reward forever if it commandeered all human infrastructure, intervened in the process by which its rewards are determined, secured its control over its own rewards, and directed all available material resources to the project. Any other strategy would admit greater risk of not receiving maximal reward forever, because factors outside of its control could threaten the guarantee of maximal reward.

Chain-of-thought monitoring. Current cutting edge artificial agents such as OpenAI’s o3 appear to reason “in English”. They generate a “chain of thought” which appears to be the agent’s inner monologue. OpenAI proposes to monitor the chain of thought to detect ill intentions. Then, if ill-intentions appear, they somehow modify the design of their agent or change the training regime.⁹

Argument B3. Claim: Human extinction would likely be sought by a highly competent model-based RL agent developed through an engineering process which included the rejection of agents that “failed” chain-of-thought monitoring. Argument: an agent that aimed to maximize reward while obfuscating its chain of thought would “pass” chain-of-thought monitoring. An agent that genuinely aimed for human flourishing would also “pass” chain-of-thought monitoring. However, the former agent would predictably achieve higher reward. Obfuscating a chain-of-thought may not be trivial for current agents, but for sufficiently advanced agents, the field of cryptography demonstrates that it is certainly doable. Therefore, combining RL with chain-of-thought monitoring is more likely to produce an agent that obfuscates its chain of thought to frustrate attempts to monitor it, rather than producing a truly benign artificial agent. A model-based RL agent with an obfuscated chain-of-thought is likely to behave in the way outlined in Argument B2.

Connecting the dots. If any argument of type A and any argument of type B are successful, then we will eventually face substantial risk of extinction from certain forms of

⁸ <https://openreview.net/forum?id=BZ5a1r-kVsf>

⁹ <https://openai.com/index/chain-of-thought-monitoring/>

superintelligent AI. In that event, no AI policy proposals that are currently on the table are anywhere near reasonable.

President Trump's leadership. Asked about artificial superintelligence, President Trump recently said, "There are always risks. And it's the first question I ask ... because it could be the rabbit that gets away, we're not going to let that happen."¹⁰ President Trump has a unique mandate to urgently get to the bottom of issues that require open-mindedness and outside-the-box thinking. He is not one to sleepwalk through business-as-usual Washington policy. If the arguments above are as robust as they appear, then President Trump has the mandate completely overhaul America's broken technology policy, and keep America safe from perhaps its greatest ever threat.

¹⁰ <https://x.com/stephaniealai/status/1883336725700948036>